

RUNNING HEAD: videogames literacy

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Introduction

Over the past decade, the popular press, media scholars and now educators have been paying closer attention to computer and videogames (hereafter referred to as videogames) (Herz, 1997, Cassell & Jenkins, 1998; Pool, 2001). Creatively, videogames push the boundaries of interactivity, immersive environments, community design, and digital storytelling. Technologically, games push the boundaries of consumer-grade real time simulation and artificial intelligence. Culturally, they are changing the way we play, learn, and interact, quintessential sites of broader shifts in knowledge consumption and production (de Castell and Jenson 2006; King, 2001; Scholder & Zimmerman, 2003; Squire, 2003). Games are in a transitional phase of cultural status, and it is no longer unusual to see game exhibits at art museums, or university courses on gaming. The last 18 months have brought us dozens of academic, government, and industry conferences focused on the academic study of games, a substantial portion of which are dedicated to games and learning. It is beyond the scope of this chapter to identify all the major conferences related to games currently running each year, but noteworthy U.S. industry conferences include the biannual Serious Games summits held in Washington D.C. and in conjunction with the Game Developer's Conference, and the Education Arcade, held in conjunction with the Electronic Entertainment Exposition. Although the latter was founded in the United States, similar conferences are or have been held in Europe, Japan, and Australia. Academically, there are at least a dozen relevant conferences every year including the regular meeting of the Digital Games Research Association (DiGRA), and the University of Wisconsin-Madison's annual Games, Learning, and Society Conference. The Games and Learning research centers at the Massachusetts Institute of Technology, the University of Southern California, the University of Texas-Austin, and the University of Wisconsin-Madison are becoming more established, and are formalizing into degree programs at the undergraduate, master's and doctoral level (for more on such developments within academia, see Carlson, 2003, and Schiesel, 2005).

Games are one – perhaps the *quintessential* – site for studying digital literacies as a medium of *interactivity*, both with respect to the human-computer interface and among constellations of users. Videogames are the medium of the computer, and in understanding them we can understand what it means to think, act, and learn in simulated worlds (cf., Starr, 1994). However, researchers are only beginning to theorize how games operate and understand what the medium will mean for learning, schooling, and society. Key questions that currently shape this growing field of studies include: What are the implications of an interactive medium for literacy? How does access to globally distributed communities change traditional notions of meaning making? How will a gaming generation react to traditional schooling with its grammar of teacher / text as authority and student as product, when outside school this generation relives historical eras, lead civilizations, conduct forensic investigations, or can earn a real wage by buying and

selling virtual currencies? (Castronova, 2001; Shaffer, Squire, Halverson, & Gee, 2005; Squire, 2003; Steinkuehler, 2004a; b, and this volume).

This review of gaming research begins by examining the study of *games as artifacts*, and discusses the assumptions academics have made about and approaches they have used to study games. This section is followed by an examination of studies of *game playing as a social practice*. This focus encompasses research that investigates gaming in communities and social contexts. The final section explores the implications of games for literacy, learning, schooling, and education writ large, focusing particularly on studies of games in educational contexts. Across all three sections I argue that games are an experiential, interactive medium where we participate (and co-create) new worlds. Although these worlds are synthetic, simulated worlds, they are worlds constructed to provide particular kinds of experiences, which might be called designed experiences. Games literacy can be defined as developing *expertise* in designing rewarding experiences for oneself within a gameworld (particularly within the game's semiotic and rule systems). Notably, there is no one rubric by which expertise can be judged, and through studies of gaming communities we see how different meanings are created, negotiated and given legitimacy. Like video games themselves, gaming communities are interactive in nature, frequently functioning as spaces for participating in social practices with a life beyond the boundaries of the game itself. To date, efforts to create such programs in schools have shown some successes, but fallen short of creating the kind of participatory culture that characterizes the contemporary popular culture landscape.

Games as Artifacts: Synthetic Spaces for Action

Videogames' interactivity presents major challenges for literacy researchers. This includes asking how we best can treat authorship in a medium in which players have *agency* in creating the unfolding of events and in changing the very symbols and representations that comprise the game on screen (Jenkins, in press; LeBlanc, 1999/2006; Robison, 2005). In other words, without the player there literally is no game. This leads to a core challenge for literacy researchers: How do we study a phenomenon that cannot exist without its players? For example, in the case of *Grand Theft Auto: San Andreas*, a first-person shooter game, the much publicized example of a player stealing a car, hiring a prostitute, and running her over is an emergent property of interacting rules; however, most players may never explore this aspect of the game, and certainly would not use it as a regular strategy. Choosing a single set of interactions in a game for scrutiny is unproductive and disingenuous although it is a common tact of mainstream media reporting of games. Yet, clearly there is a difference between *Grand Theft Auto* and *Sim City*. Just how to categorize, conceptualize, and understand the characteristics of games and particularly has been a core concern in games and literacy research.

Perhaps not surprisingly, then, much of the early (and current) theoretical work in videogame studies has aimed at defining what a game is and how games ought to be studied. The core organizing debates have centered on whether games should be studied from the perspectives of media studies (this orientation has been characterized as "narratology") or as a form of play (this orientation is often called "ludology") (cf., Frasca, 1999). These theoretical issues take on practical significance as researchers deal with questions concerning violence and media in games like *Grand Theft Auto: San Andreas*. While media theorists tend to emphasize the importance of

characters in a game and the narrative (and rule-based) consequences of game actions, ludologists tend to emphasize the psycho-social dimensions of play (cf., Wadrip-Fruin & Harrigan, 2004; Frasca, 2003).

The so-called narratology / ludology debates have been unfortunate, since the interaction between “narratology” (taken here to mean characters, backstory, symbols) and “play” (fantasy, removal of every day rules and constraints, goal-driven activity) is just what makes games so intriguing (Juul, in press; Squire, 2005; Squire & DeVane, 2006). To continue with the case of *Grand Theft Auto: San Andreas*, part of its playful and theoretical allure is that it is a stylized representation of a “ghetto;” a representation that codifies symbols like gang colors and allegiance into game concepts like loyalty and respect, giving players a set of symbols and icons to play with. Reports of research focusing on players suggest that gamers do wildly different things with the worlds available to them in this game and, in fact, appear to conceptualize the worlds and themselves within them according to their own lived histories (Squire & Devane, 2006; Vargas, 2005). One player interviewed by Squire and Devane used the game as little more than a “pimp your ride” car customization kit (he wanted to pursue a career in car customization), one used the game as a chase scene simulator as a relief from school, and another as fantastical escape from the suburbs. Thus, both technologically and conceptually, it is the (usually) seamless blending of agency, interactivity, narrative, semiotic domains, and expansive worlds that makes video games so interesting (Gee, 2003; Mateas, 2004; Murray, 2004; Perlin, 2004; Poole, 2000).

Games as spaces

Jenkins (1993, 2004) argues that one useful way to understand games-as-interactive objects is to study them as spaces. Writing about Super Nintendo’s *Mario* games, Henry Jenkins and Mary Fuller (1995) argue that the narrative structure of videogames is like those of travel narratives. A game like *SuperMario* is organized around exploration; the core game pleasures involve visiting exotic lands, meeting unusual characters, and unlocking secret passages. Jenkins and Fuller argue that structurally, this kind of exploration game shares much in common with John Smith’s travel writing:

Nintendo®’s central feature is its constant presentation of spectacular spaces (or “worlds,” to use the game parlance). Its landscapes dwarf characters who serve, in turn, primarily as vehicles for players to move through these remarkable places. Once immersed in playing, we don’t really care whether we rescue Princess Toadstool or not; all that matters is staying alive long enough to move between levels, to see what spectacle awaits us on the next screen. Most of the criteria by which we might judge a classically constructed narrative fall by the wayside when we look at these games as storytelling systems. In Nintendo®’s narratives, characters play a minimal role, displaying traits that are largely capacities for action: fighting skills, modes of transportation, pre-established goals (1995, n.p.).

Thus, while many cultural critics may look at Nintendo’s games and see impoverished potential for interaction, Jenkins and Fuller remind us that not all “classic” literature is

as taught in Freshman English. Different narrative forms have served different purposes and taken on different forms and interpretations across different times. For example, in the early 1700s, most contemporary novels were regarded as of dubious quality, or, as in the case of Daniel Defoe's *Robinson Crusoe* (1719) – which was described as a “true history” and not a novel – were seen to have little value as a resource to be used in educating children.

For Jenkins and Fuller the task for games scholars is not to argue *how* games necessarily fit within traditional forms, but to better understand how games are *transforming* previous forms. Taking (and fueling) the narratology / ludology debate, Jenkins (2004) argues that many games do in fact aspire to tell stories, but they do so by means of “space.” Jenkins sees a basic contradiction between the linearity of traditional narrative structures and interactivity, which leads designers to telling stories through the arrangement of space so as to produce compelling narrative moments – or *experiences*. Jenkins finds four ways that games use space to create narrative: evocative stories (stories that reshape based on previous characters and media), enacted stories (stories where players enact specific plot points) embedded narratives (games where narrative events are embedded in artifacts in the world), and emergent narratives (narrative experiences that emerge as the result of simulation). Jenkins situates his argument for understanding stories and games within a *pragmatic* framework that tries to account for how games are designed and played, drawing on interviews with game designers as well as a close read of videogames.

In “The Art of Contested Spaces,” Jenkins and Squire (2002) expand on this notion of understanding games through spatial terms, describing different aesthetic approaches for creating *conflict* within space. Responding to critics who want to see a theory of digital games rooted in play and games (e.g., Eskelinen, 2004), Jenkins and Squire describe how *contestation* is at the core of game design. Many videogames, such as *Civilization*, remediate board games, while others, such as sports games, are about contestations of space on a metaphoric or strategic level (e.g. a basketball team that uses floor spacing well or defends the lane well). Enemies, puzzles, and a character's inability to move or affect the world obstruct access to space. Much of game play can be seen as “opening” new spaces – both physical ones (levels) and abstract ones (possibilities, or new experiences within the same physical space). Jenkins and Squire also emphasize the social space surrounding the machine, arguing that interfaces and game mechanics from the guild user interfaces in the medieval-like *Asheron's Call* to the hand-held maracas in the music game *Samba de Amigo* are designed to promote social interaction. A critical feature of these spaces for Jenkins and Squire is that they evolve in response to users' actions. In role playing video games like *Morrowind*, players' actions open new lands, social networks, and possibilities for action (cf., Hayes, 2005; Kadakia, 2005). In snowboarding games, the camera, snow, and crowds all respond with exaggerated feedback on players' actions (e.g., crowds will cheer or moan according to the quality of the gamer's turn, slide, jump etc.; the camera follows the player as she makes decisions about which downhill run to take and how to complete it). Or, perhaps most dramatically, in *Black & White*, the physical features of the world literally evolve in response to players' actions, potentially becoming everything from a tropical paradise to a barren desert, making the world a physical manifestation of players' actions.

Game Time: Games as dynamic systems

Games are dynamic systems operating similarly to simulations, beginning with a particular set of initial conditions and then evolving through time in response to players' actions. Of particular interest to game theorists has been how game players report transformations of their experience of time while engrossed in game play. Csikszentmihalyi (1990) describes this phenomenon in terms of flow; the condition under which a player's skill level is optimally matched with the level of challenge faced, creating conditions in which the participant reports a feeling of being in the zone. Researchers like Bowman (1991) have shown how videogames are especially good at inducing and maintaining these flow states by adjusting complexity in response to users' actions. Myers (1991; 1992) seeks to extend this hypothesis concerning play and time, and posits that game play unfolds through *semiotic transformations*; transformations where players' goals and oppositional structures iteratively evolve, resulting in an emergent, semiotically more complex self over time.

More recently, some researchers have begun to use *time* itself as an analytic axis when examining games and game playing. Jesper Juul's project (2004), for example, has been to build a descriptive theory of digital games that uses time as one lens for examining game aesthetics. Juul is especially interested in how players interact with the *game state* over time. Juul uses the term, "game state," to describe how the "game" may be static code on a computer, but as game play unfolds dynamically, the game and player together generate unique game situations. *Sim City*, for example, consists of no actual city, but rather a virtual parcel of land, a set of nouns (such as land, road), a set of actions (such as zoning), a rule set describing relations among these things, and the ability to watch these relations grow and evolve over time. Juul categorizes games in terms of how game *play time* (time experienced by the player) relates to *game time* (whatever chronological system is employed in the game). Juul's work suggests that there is no one "static" game to be studied, only game *states* that emerge through the interaction of preset rules and player choices. Indeed, replaying a video game never unfolds exactly the same way as it did originally; therefore, issues concerning author, audience, and meaning making are only amplified in game studies (Church, 2001; LeBlanc, 1999/2006; Robison, 2005).

Interactive "texts:" This lack of a stable "text" poses one of the greatest challenges for educators. Designers of educational games, like Aldrich (2004), have observed that the single biggest obstacle to creating educational games is moving from linear to dynamic content. Aldrich's (2004) model for designing educational games is built around *cycles of action* (cf., Games-to-Teach Team, 2003; Salen & Zimmerman, 2002; Squire, 2005a - e). These cycles of action consist of feedback loops whereby the player iteratively develops goals, takes an action, experiences feedback on that action, and shifts goals and actions accordingly. Similarly, Betrus's (1995; 2005) chainsaw model is another time-based approach to understanding games that emphasizes the cyclical, recurring nature of game play (see Figure 1). Betrus' model – which is particularly appropriate for action games – focuses on the relationship between relatively short events (e.g., moves), sets of moves (e.g., sequences), and entire levels. Betrus uses this model to examine the ways in which these elements are linked together by means of a theory of how they unfold over time. For Betrus, games are marked by small, repeating challenges that are punctuated by more difficult challenges that require a consolidation of previous skills (See Figure 1). This chainsaw model suggests how games are designed for learning; the player begins

with small, relatively easy challenges that are paced to produce ‘automaticity’. These regular events are punctuated with more challenging sequences which serve to both disrupt and then strengthen or add to players’ knowledge (cf., Gee, 2005).

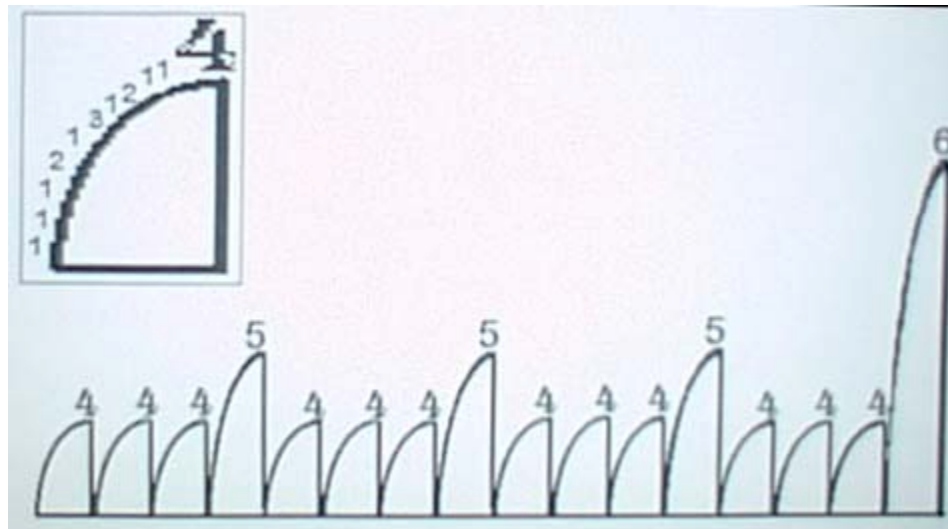


Figure 1: A screenshot of Betrus’ Chainsaw model. (Image used with permission from the author).

Good games build on themselves by taking these smaller units and combining them to create increasingly complex situations that coalesce to become a holistic experience for the player-as-game-character. Gee (2004; see Sheffield 2005 for a description), examines *Ninja Gaiden* as one such game that is designed to produce mastery. Gee shows how the first hour of *Ninja Gaiden* is constructed as a *de facto* tutorial, teaching the player how to read the game environment, perform isolated skills, perform sequenced skills, and finally, produce mastery by the first boss battle. Gee argues that *Ninja Gaiden*, when taken as a piece of instruction is artfully constructed to produce learning, learning that is not just of any sort, but in this case, teaches the player to play like a ninja (as opposed to, say, like Rambo).

Cycles of Choices and Consequences. Games are driven by these overlapping, interacting cycles of action; as legendary game designer Sid Meier has widely been quote as saying, “Games are a series of interesting choices” (Meier n.d.). Most any game – from turn-based strategy games like *Civilization* and massively multiplayer games like *World of Warcraft* (worldofwarcraft.com) – can be analyzed in terms of overlapping cycles of choices and consequences (Salen & Zimmerman, 2003). Games like *World of Warcraft* excel at just this type of play phenomenon; players are constantly developing overlapping goals, so that a player begins a quest with the idea of earning experience points, but midway decides to gather materials needed to create a new weapon, as well. Once the quest is completed, she might have nearly enough raw materials (say, a certain amount of copper) to craft a new weapon, so she keeps playing in order to gather what she needs in order to meet her “new weapon” goal. Just before she gathers enough copper, however, she discovers a new land that she wants to explore. These overlapping cycles of choices and consequences are in part what make games so “addictive.” In *Civilization*, as another example, players report a “just one more turn” phenomenon. The player develops a web of interlocking goals that overlap in such a fashion that she stays playing for

hours on end, hoping to experience the results of one set of decisions (e.g., completing a quest to obtain new equipment). In the meantime, however, she becomes engrossed in experiencing another (e.g., collecting flowers to raise herbalism skills). In this fashion, many good games are specifically designed so that players always feel as if they are on the verge of “doing something interesting.” Roughly translated, this means being put in a novel situation with the opportunity to think creatively while solving a problem (McKenzie, 2005; Shaefer, 2004). This need to keep the player in a constant state of “nearly doing something interesting” is what drives game designers to constantly iterate and reiterate game designs through play-testing, so as to achieve a fluid game experience for the player.

Games Remediating Music. Emerging from this work on games and time is a sense that games are a deeply *rhythmic* medium that may share as much with music and dance as they do with visual storytelling. This is not to suggest that there is no rhythm in storytelling (oral storytelling is especially rhythmic, for example), film editing, visual composition or theater. It does, however, persuade us to go beyond games’ graphics and consider them as experiences that are fundamentally multi-modal (cf., Lemke, 1998).

Squire (2001) and Smith and Squire (2002) connect games with a tradition largely overlooked in game studies: virtual soundscapes. Building on Jenkins’ notion of narrative architecture, Squire and Smith describe games as a sort of *sonic architecture*, operating much like music which, in the words of Ong (2002), places the listener at the center of the world. Like music, the aesthetics of many games is less about the unfolding of a plot than it is about the *feeling of being* somewhere. Games from *Samba de Amigo* to *Quake* operate through an aesthetics of *rhythm* and a feeling of empathy with the game system (c.f. Gee, 2005). For example, in *Arcadia*, a web-based game developed by gamelab, players play simplified versions of four “classic” games simultaneously. Each of which has its own distinct rhythm, producing a polyrhythmic sensation in the player. Friedman (1999) argues that a primary pleasure of playing *Civilization* is entering the system, developing an almost cyborg relationship with the computer whereby the game world is one that we inhabit via the game controls which, in turn, become an extension of the self. Even though *Civilization* is a turn-based game its turns and overlapping cycles of interaction have their own cycles and rhythms, producing a particular kind of feeling (which, notably, the designers have sped up in the current version, *Civilization IV*). Gee (2005) and Jenkins, Squire and Tan (2004) have argued that designing video games where we can enter physical, biological or social systems is a promising future for the design of educational games. A case in point is *Supercharged!*, a game developed at MIT, where players can literally enter the world of electromagnetism, thereby gaining a sense for how scientists think.

Not coincidentally, the holy grail for many game designers has been to create “interactive jazz,” whereby players can spontaneously act and react in meaningful ways with the computer (or one another via the computer). By giving the player access to the controller, games place the player as both a listener and a participant in the orchestra. Indeed, at first glance players may seem to be largely constrained by game interfaces, but examinations of rhythm / action / dance players show how players use the space outside of the game to be creative, constructing elaborate dance moves with their hands, arms, and heads, treating the game as a *performative* space. On at least some level to be literate with games means being competent, if not good, with them, as

games are ultimately something one *does*. When we study games, on some level we are also studying the game performance.

Game as Performance. Fundamental to any notion of a game-based literacy is that the player is actively engaged in producing not just meanings but also tangible actions on screen and in real spaces. To be literate in the gaming medium means to be able to do things with them; one cannot imagine claiming to be “literate” with games without ever having finished a game (or a substantial portion thereof). Squire (2005a) notes that this emphasis on performance permeates much of gamer discourse, drawing on analyses of the forum-listed Frequently Asked Questions (FAQ) for *Viewtiful Joe* to argue that underlying many games’ community discourses is a *functional epistemology*, whereby *doing* in the game world organizes all accompanying literacy activities (game cheats, FAQs and discussion). Gee (2003; 2004; 2005a) contrasts this notion of literacy with the ones found in school; imagine a science classroom where students first do scientific investigations and then read books, research cheats, and discuss with peers to aid their “doing” science. Games push our theoretical notions of learning and literacy, firmly unseparating knowing from doing.

As evidence for this performative, functionalist nature of games, Clinton (2004) (analyzing *Prince of Persia*) notes that the first thing gamers do when picking up a new game is to press different buttons on the game controller (e.g., console control or computer joystick), seeing what each character (or other player role) can *do* in the world, because it is through what we *do* in game worlds that we come to know them. Also analyzing *Prince of Persia*, Davidson (2005) describes this opening portion of game play as *involvement*, where the player is generally involved in the unfolding of action but is yet to be either immersed, invested, or really controlling the action in a meaningful way. Through the careful, ever increasing orchestration of obstacles, narrative events, rewards, and new capacities, games usher the player into a state of immersion and eventual completion. Davidson observes that the final stage (perhaps most interesting for educators) is one of *investment*, where the player has been immersed in the game world, has committed to being the character for a sustained period of time.

Transformations over Time. Gee (2005b) argues that it is these transformations into “player as game player” (e.g., “James Paul Gee as Lara Croft”) that are of most interest to educators. As discussed earlier, the timescales of video games are themselves interesting, and are tied to a player’s investment in the game. A board game (like a book or film) might be played in a few hours; however, a common commercial single player video game such as *Viewtiful Joe* is usually played over a number weeks. Strategy games like *Civilization* or action games like *CounterStrike* are more akin to hobbies or sports, and are played over months and even years (cf., Wright, 2001). Similarly, the average massively multiplayer gamer plays any one title for 6 months to a year, on an average for 20 hours a week (see Steinkuehler, this volume). Timescales may be a production notion for educators who seek to understand the transformations that occur through game play, as they suggest how different game experiences might create different learning trajectories (Lemke, 2005; Squire, 2005).

Indeed, Lemke (2004) argues that games are especially interesting laboratories because they allow us to examine learning over timescales, zooming from the micro action to macro participation in communities of practices. As we shift the timescales of game activities we see

different trajectories of participation. On the micro-second level, we can analyze a specific move within a game environment (such as a kick or punch in a fighting game). Building on Lemke's (2000) framework of learning over time scales, we can see how any particular game practice (such as a executing a kick or punch) forms through the affordances of actions occurring at smaller intervals timescales (visual perception of data in the environment, muscular movements manipulating the controller), as well as broader encompassing systems (the duration of a battle or fight). Videogames are intriguing combinations of closed and open systems; game systems are ultimately hard coded and knowable, but their emergent properties are so complex that they are for all practical purposes open-ended problems (Steinkuehler, 2005).

Summary. Video games are constructed worlds, worlds that are built according to particular sets of rules that allow certain types of action. As such, they are *ideological spaces*; worlds constructed by particular viewpoints that tend to lead toward the expression of particular ideas (Squire, in press). However, they also are participatory in that they invite players to *inhabit* characters that have particular capacities in those worlds and that are designed to elicit particular performances or “ways of being”. These performances are situated within communities of players that also remediate the game experience. As an example, compare different groups of *Quake* players. Highly competitive *Quake* players consider the game a competitive match of skills, even turning down (or off) graphic features to maintain a competitive advantage by speeding up their frame rates (the rate at which the computer displays the game state). Casual players may enjoy the graphics and character designs, whereas machinima players – those who use the game engine to create movies – don't particularly care about the gameplay at all. Indeed, different game communities construct gaming very differently, and have competing ways of framing game play, which may eventually stabilize into accepted systems for “reading” the technology (c.f. Bijker, 1995). At the current moment, however, games are noteworthy for their interpretive flexibility, with different game meanings tied to different game communities.

Game communities

Literacies are enacted and legitimized through participation in communities (Gee, 1990; The New London Group, 2000). As with any literacy, it is not enough to generate meanings from texts; interpretation occurs within communities that legitimize particular perspectives and ways of knowing while marginalizing others. As Squire (2005c) points out, we see this same phenomenon in game communities, the meanings of games are argued and debated in various communities, with each interpretive community generating its own criteria for what constitutes a compelling game, how games ought to be read, and what a game such as *Grand Theft Auto* is “about” (Weise, 2004). These arguments range from whether “story” is a legitimate ground for critiquing a game, or the extent to which game sequels remain faithful to the original property. Critiquing texts without considering the interpretive frames in which they are studied / used is problematic in any context because it privileges the interpretive systems of academics, cultural critics, and politicians over those who have an investment in the media itself. In short, studying games as texts without taking into consideration what players actually *do* with them is problematic for a number of reasons. Some of these are discussed below.

Mirroring the history of media research more broadly, game scholars have been relatively slow to study games communities, first studying game boxes, then games, and not until recently, actual people playing games (Provenzo, 1991, 1992; Squire, in press). Perhaps for obvious reasons, the most comprehensive and well studied examples of gaming communities to date are of massively multiplayer gaming space (c.f. Steinkuehler, 2004b; this volume). Massively multiplayer games feature thousands of people interacting in real time from around the globe. Although massively multiplayer online games are not the focus of this review, research on these games connects to research on more traditional gaming communities in at least two major ways. First, guilds or clans – a key community unit within massively multiplayer online games – are a social structure that frequently span and cross both massively multiplayer online games and traditional games (Steinkuehler, 2005; Jakobsson & Taylor, 2003). Second, game literacy practices that characterize massively multiplayer online game play are frequently core to single player games as well. For example, in Sweden, where the video game CounterStrike is the third most popular sport, companies who sell and maintain server software that support game communities report that players spend 4 hours using community features (chatting, participating in forums, reviewing film clips) for every one hour they spend playing (Squire, 2005).

Studies of gamer communities suggest the importance of not just focusing on games, but looking at gamer *discourse* to understand emerging digital literacies. Thus far, much of the academic work on games literacy has focused on the construction of games more than the social interactions surrounding them, which is understandable given that games are the leading consumer applications of artificial intelligence, real time simulation, and interactive narrative. Studies of games and their attendant social structures suggest, however, that although digital literacies are driven by technological shifts, they are instantiated and then promulgated through social institutions and structures such as guilds, clans, forums, affinity spaces, and self-organizing learning systems like Apolyton University (c.f. Squire & Steinkuehler, 2005). These sites are critical expressions of games culture, sites where gamer discourse is formed and maintained.

Games as liminal spaces. Talmadge Wright and colleagues' studies of *Half Life* and *Counterstrike* communities is one of the longest running studies of gamer culture (Wright, 2002; Wright, Bouria, & Breidenbach, 2002). *Half Life* is one of the most commercially successful and critically acclaimed video games of all time. *Counterstrike* is a fan-produced modification for the game that has attracted a dedicated following of millions of players, and for years was the dominant competitive multiplayer squad-based shooter game. As Wright (2002) notes, at first glance *Counterstrike* is an incredibly violent game with juvenile discourse patterns. Wright and his colleagues' studies reveal not anarchy but, rather, a complex social world with rules and social conventions organized around an ethos of "having fun." This core feature of *Counterstrike* communities is constituted through creative player actions, which include players' verbal banter and one-up-manship, creative reappropriation of game actions (such as committing suicide in a humorous manner to amuse one's peers), and production of original art that can be viewed in-game. Wright et al. (2002) argues that the meaning of *Counterstrike* is generated by its players who have created a set of mores that are socially reproduced and handed down from one generation to the next. Player *production* is at the core of this community, and the ability to entertain and amuse is held in highest regard.

Player production was also the focus of Zhan Li's (2003) Master's thesis research: *The Potential of America's Army as Civilian Public Sphere*. *America's Army* is the controversial \$8 million video game / recruiting tool built and distributed by the U.S. Army. The goals of the project were to "support Army recruiting efforts, particularly of teenagers with high-tech aptitude and skills; raise the positive profile of the Army as an interesting, high skilled organization; and to promote the revival of military-civilian grassroots contact" (Li, 2003, p. 8). Li's ethnography examined how the activities of three exceptional gamer groups (real life soldiers and veterans; Evangelical Christians; and hackers) intersected with the stated and unstated purposes of the game.

Li finds that *America's Army* functions as neither propaganda for the state, nor as a site where hacker culture wholly subverts the culture of the game but, rather, as a (contested) *liminal* space that where everyday relations can be (and are) reconfigured for the purposes of play. Li's notion of liminality differs somewhat from Wright and colleagues' (2001), in that whereas Wright's notion of liminality is largely about activity within the game space, Li relates situates players activities within other social spheres (i.e. military service, religious). Even though the formal game play elements of *Counterstrike* and *America's Army* are essentially the same, the forms of play that unfolded in each was quite different. Li argues that *America's Army* functioned as a space where the "rationalities" of the state, military, commodity, and people's lived experiences are in collision. For example, Li observed that many players stayed away from *America's Army* during the initial phases of Gulf War 2, wanting to preserve the "gameness" of the space; that is, they did not want the seriousness of the actual war to be compromised by "playing at war." Similarly, Li describes soldiers and veterans gamers in the *America's Army* community as soldiers acting at playing soldiers (2003). That is, these soldiers and war veterans were using *America's Army* as a context for inhabiting stylized identities as soldiers. Li uses the lens of Army, Christian, and hacker groups to show how the space of *America's Army* is a peculiar mix of both the intentions of its designers (a military-owned space for civilian / military relations) and the ideologies of its inhabitants.

These studies suggest that games' most potent social value may be their *liminality*, their capacity to function as contexts within which participants can play with new identities and ideologies. As role playing game designer Warren Spector (2001) has argued (building on the work of Victor Turner), every culture has provided spaces where the dominant social order is temporarily reshaped, and people have (at least temporarily) the ability to pursue experiences with reduced (or removed) social consequence. Li's study suggests how games are co-constructions of the designer and participant, in that emergent activity reflects the goals of the participant constrained within the parameters of the game spaces which are the residue of designers' goals. In *America's Army*, for example, it matters greatly that the representations, symbols, game rules, social rules, and capacities for action are based on those of the military. As researchers, however, we know little about their meanings until we examine how they are taken up by players.

Games as Affinity Spaces for Learning. James Paul Gee (2003; 2004) argues that videogames are powerful models for learning. We can examine the technical design of games, how games are *designed as experiences* to produce certain feelings, skills, and knowledge; games can also be studied as *social spaces*; spaces where people congregate to learn. Examining forums for *Age of*

Mythology, Gee (2004) proposes that games are affinity spaces, spaces where learners congregate based on personal interests rather than race, class, economic status, or background. Gee describes several key features of such forums: (1) participation is open to any user; (2) newbies and veterans share a common space; (3) participants have the power to shape and transform the environment; (4) knowledge and expertise is distributed across players; (5) multiple routes for participation, success, and status are presented; and (6) participation in the affinity space reshapes the “attractor” (i.e. game) itself. Gee contrasts these affinity spaces with communities of practice, arguing that affinity spaces are much more common to today’s digitally mediated society where technologies makes social boundaries permeable, participation open, and knowledge production decentralized.

As such, affinity spaces are more than just intriguing places to spend our off hours; they are exemplars of the social organization in the modern digital world that are changing contemporary notions of literacy. Gee (2004) connects the rise of affinity spaces to broader patterns in the new capitalist world (cf., Squire, 2004); others (e.g., Jenkins, in press) have described similar trends cutting across entertainment, political, and educational media. Today’s digital consumer doesn’t have a stereo or piano; they have a suite of digital tools where they can “rip, burn, mix,” and then publish their creations via the web, using digital services to even turn a profit if their work is good. Core to such affinity spaces are opportunities and expectations for *participation*. The rise of digital production and communication tools make participants producers and not just consumers of media, and games are a leading technology for such changes (Squire & Steinkuehler, 2005).

Self Organizing Communities for Learning. These studies of affinity spaces beg the question of how to design learning systems for a digital age. Although substantial learning occurs in such affinity spaces, they are not designed explicitly for learning. What might a game-based learning system built on these principles look like? To better understand this, Squire and Giovanetto (in press) have been examining Apolyton University, an online community of game players that has evolved to teach other players to become expert gamers. Squire and Giovanetto argue that unlike most other institutions of higher education, the system is driven by *learning*, with class practices, courses, and administrative structures all taking a backseat to the demands of learning. The core practices driving the community are playing *Civilization III* and a) redesigning a common game modification to improve its playability; b) conducting During Action Reports that capture participants’ cognitive activity and serve as the basis for discussions, and c) proposing and developing new “courses” for the community. As players create and change courses, they create custom game modifications to illustrate their arguments about the game, and collaboratively build a “master” curriculum / game file.

Squire and Giovanetto (in press) argue that participating in Apolyton University is a process of transitioning from a game player to a game producer. Players enroll as advanced players, having spent dozens, if not hundreds of hours with the game and having mastered its basic rules. As players begin to identify and exploit loopholes, they propose and implement changes to the games’ rules, identify superior strategies, and invent new game rule systems, including custom modifications and scenarios. In one course, lead artificial intelligence designer for the game, Soren Johnson, joined the discussion to correct a misconception about how the game’s artificial intelligence works. As a result of the ensuing discussion, several players from

the community were recruited to participate in the game play testing and balancing, completing the cycle from game consumers to producers.

Self-organizing learning environments are unlike traditional online learning environments in that they are open systems with permeable boundaries (Squire & Giovanetto, in press; Wiley & Edwards, 2002); participation by participation is open to anyone and the “classroom” activities are both transparent to outsiders and expected to have an impact on broader social practice. In the case of self-organizing game communities like Apolyton University (there are numerous others, including Civfanatics, Madden Football forums, or specific sites within the Warcry network), sites of learning spontaneously organize around authentic, meaningful questions and typically result in changing the design practice, effectively reshaping the context of their learning (Gee would call this “reshaping the attractor”). As such, self-organizing learning communities share much more in common with cutting-edge research communities than with formal classrooms. Their existence suggests an implicit critique of schools: Why should I study computer science or writing classes in order to get a job in the entertainment industry, when I can participate in learning communities that offer trajectories of participation where I might enter the industry? Game-based self-organizing learning systems are intriguing to educators for how they are at the cusp of participatory media culture; in today’s digitally-mediated media environment, meaningful participation involves opportunities for participating in the *production* of media. These examples also suggest design patterns for the design of game-based learning environments, which will be discussed in the final section.

The consequences of participation in gaming practice for literacies. Studies of digital gaming communities as liminal spaces (e.g., Li 2003), affinity spaces (e.g., Gee 2004), and self-organizing learning systems (e.g., Squire & Giovanetto, in press), suggest how games are more than a technology; they are a maturing medium with attendant social practices. The preceding studies suggest profound shifts in participants’ relation to knowledge, media, information, social institutions (c.f. Squire & Steinkuehler, 2005). These practices include (1) *producing* as well as consuming information, (2) the strategic use and critique of digital spaces that promote their particular values, (3) the ability to produce meanings across multiple representational forms (i.e. texts, visuals graphs, charts, game modifications), (4) texts that circulate within and across communities, (5) trajectories for participation in social systems, including game journalism, design, and criticism, and (6) the repositioning of written texts from objects of authority to resources that are used in support of (digitally mediated) practice. The lasting social consequences of these shifting practices are only beginning to be understood.

Preliminary survey work on the games generation suggests that these discourses are powerful ones, shaping our notions of learning, literacy, and expertise. Surveys of gamers show that they have an increased appetite for risk, greater comfort with failure, a stronger desire for social affiliations, a preference for challenges, capacity for independent problem solving, and a desire to be involved in meaningful work when compared with non-gamers (Beck & Wade, 2004). Underlying Beck and Wade’s argument is a notion of changing literacies; gamers have grown up with a medium built on assumptions unlike those in print cultures (e.g., a game engine can be tinkered with, a text is not necessarily print-based or defined by book covers); game players are co-authors along with game designers, co constructing the game-as-text through their own action (c.f. Robison, 2005). Gamers have grown up in simulated worlds, worlds where

anything is possible, and where learning through trial and error is expected, information is a resource for action, and expertise is enacted through both independent and collaborative problem solving in self-directed tasks (Simpson, 2005).

How students will react to traditional schooling in this digital era is a question of increasing importance that digital literacy scholars are beginning to ask. Kevin Leander and colleagues investigate the in and out of school literacy practices of one gaming youth, finding that texts and literacies operate quite differently in and across these contexts, and actually serve to bring down traditional home/school conceptual divides in education research (Leander 2005; in press). Using Actor Network Theory, Leander and Lovvorn (in press) find that multi-modal texts in gaming environments function as resources for action. These texts come from official and non-official sources and different media (strategy books, websites, in game text and non-text resources) (c.f. Squire & Steinkuehler, in press; Steinkuehler, 2005). Within a game environment, the meaning of texts (and their legitimacy) arises in response to their efficacy for action. Leander and Lovvorn contrast this type of meaning in texts with those in classrooms, where texts are given authority by their *distance* from the student; the teacher's notes (wherever they came from) and published works are given the most authority, and students' work is to effectively reproduce those notes for legitimized authority. In contrast, texts in game cultures are "lively," flowing across boundaries and borders as they serve social functions, texts in classrooms are remarkably *static*, designed for a student or teacher's consumption, but having little impact outside of the classroom. Leander and Lovvorn show how in the case of one typical student (Steve), text (and all of its associated artifacts, such as notes) are prepared and produced exclusively for a teacher's consumption, and their life ends abruptly when the teacher grades them, having no discernable impact on the field, the teacher's thinking, other students' thinking, or Steve's life.

The preceding studies of gaming cultures suggest how emerging game literacies are a product of both games-as-designed-artifacts as well as the game cultures in which they are created and enacted. Indeed, the values that are expressed in the design of games do not emerge from thin air; they are the product of particular technological cultures that spawned them, which, as Stephen Levy argues, can be traced principally to hacker cultures in terms of origin (Levy, 2001). Games are then enacted in cultural contexts, where such values are promulgated. Through sites like Apolyton University or Game FAQs, we can see examples of gamers creating social structures that initiate newcomers into these cultures. The affinity spaces that Gee (2004) describes are the result a co-evolving structure of emerging technologies (games and modification or "modding" software, The Internet, wiki software, online community software), and the communities that spawn them. If the games themselves serve as attractors, then educators may benefit through studying learners' trajectories through these spaces, focusing on how values and identity emerge in practice. To date there are still comparatively few such studies, but future literacy scholars might benefit by examining how participants learn through participation in these structures.

Games in schools

Thus far, the discussion of games and learning has focused largely on learning and literacy outside of classrooms. As Warschauer (this volume) observes, emerging literacy scholars have

generally sought to understand (1) how literacies are changing in response to emerging technologies, frequently with a focus on how traditional schools are failing to respond to these changes (2) the power structures and dynamics behind literacy regimes, and (3) how literacy can be improved (or the impact of media on learning). The incongruencies between emerging game and school based literacies has meant that necessarily, the bulk of the research on games literacies have taken place outside of classroom contexts. There are, however, examples of research programs where game literacies are examined in classroom contexts, usually occurring with a design-based research framework. Design-based research studies seek to posit and test hypotheses by creating learning contexts and then examining the processes by which they function (Barab & Squire, 2004; Cobb, Confrey, diSessa, Lehrer, & Schauble, 2004; Design-Based Research Collective, 2004). Studies of games in classroom settings suggests that game play can be the basis of meaningful academic learning experiences, but the epistemological assumptions of learning through game play are at odds with those of schooling. This suggests a need for radical transformations of schooling to meet the needs of a digital age (Squire, 2004).

Games, simulations, and play have a long history in education and the study of human learning more generally. Legendary game designer Chris Crawford (1982) remarks that games are “the original” educational technology, reminding us that most mammals learn through play – creating and playing within spaces set apart from the every day. This rhetorical framing of game – such as, play as development – is but one of many rhetorics of play. In his magnum opus, *The Ambiguity of Play*, Sutton-Smith (1997) argues for the centrality of play in human experience contends that play has been conceived primarily along seven or so rhetorical lines: the ancient four, (1) fate, (2) power, (3) communal identity, and (4) frivolity, and the modern discourses of (5) progress, (6) imaginary, and (7) the self. Sutton-Smith uses this notion of rhetorics to describe the different theoretical frames that we bring to a phenomena which then determines what we count as play (particularly how much of a context we count as play, i.e. is it an individual or an individual in context, or an entire society) and how we interpret play (what counts as potential outcomes of play) and so on. As an aside, Sutton-Smith’s work has been most closely associated with the work of the ludologists within video games research, although those associated with the “narrative” camp also draw from his work. This suggests Sutton-Smith’s work cannot be confined to a single theoretical orientation. Indeed, after a lifetime of studying play, Sutton-Smith seems to be most fascinated by the malleability of play, how psychologists, sociologists, anthropologists, and media scholars attach their own rhetorical frames or stories to play. For example, Piagetians might look at a child playing with blocks and see an individual’s cognitive development trajectory, whereas a Vygotskian might look at the broader social context of play (who gave the child the blocks, what is she building, why, and so on). Play is such an ambiguous term that the ways in which it is used and interpreted usually reveal more about the observer and their theoretical framework than the phenomenon itself (Squire, 2002; Sutton-Smith, 1997).

Although Sutton-Smith was writing about non-digital games, these same rhetorical lines also operate when digital games enter popular discourse. In particular, they are evoked when social issues around games – such as games as an educational medium or the consequences of violent game play arise. These rhetorics were evoked in the United States, for example, in the wake of the Columbine high-school massacre (cf., Jenkins, 1999; 2000). As the public searched for ways to make sense of these tragic events and debated the role that video games may or may

not have played in the teenage boys' development into mass murderers, games (particularly DOOM) were (once again) cast in socially threatening roles in order to describe how they might be affecting "gullible" youth (cf., Anderson & Bushman, 2001; Anderson & Dill, 2000; Bushman & Anderson, 2001; Gentile, Lynch, Linder, & Walsh, 2004). Indeed as Williams (2003) concludes, video games are a lightning rod for social and technological change, drawing out fear of technology and youth culture from those unfamiliar with them. Similarly, Squire (2002) argues that games are a Rorschach test for educators usually revealing as much about authors' optimistic or pessimistic view of technology and social reform as about the technology itself.

Fostering Games Literacies. One approach to fostering games literacy is to build educational programs where students develop games literacies through playing, studying, and designing games. Buckingham (2002; 2003; 2004) and Sefton-Green (2003; 2004) raise the question of how to consider games literacy in the context of the other digital / electronic / media literacies that exist (c.f. Willett, 2003; Carr, Buckingham, Burn, & Schott, 2004). Through projects like *Shared Spaces* and *Making Games*, students analyze, categorize, and understand games with digital design tools. Buckingham (2005) emphasizes critical and cultural literacies as the key literacies to foster through games. In a similar vein, Squire (2005c) raises issues concerning both what constitutes games literacies and who legitimizes what literacies (e.g., game developers, different constituencies of players, academic researchers). If one were to take the most extreme position, few outside the gaming industry – certainly few academics – would be literate in how games are designed and constructed. Further, given the interpretive flexibility of videogames, academics run into thorny issues around whose interpretations of these games are legitimized. Critical and cultural approaches to games are certainly important. To date, however, most critical approaches have reflected the norms, value systems, and sense making apparatuses of baby boomer academics rather than gamers themselves (cf., Provenzo, 1991).

Squire (2005c) argues that a fruitful approach to fostering games literacy is to use historical role playing games, such as *Pirates!*, as the focus of inquiry-based units where students develop media literacy skills. Squire used Firaxis' *Pirates!* with middle school students to investigate how games are made, how they represent history, and how they connect to other disciplines such as art, music, or computer programming. Historical role playing games take abstract issues like authorship, narrative, choices and consequences and root them in particular historical eras, making consequential issues such as representation, interactivity, and meaning. In this program, developed in conjunction with Firaxis studios, students play *Pirates!*, critique the game for historical accuracy and inaccuracy, and interact with game developers, learning how the game was made, including the particular roles (i.e. 3D programmers, AI programmers, interface programmers) available at game studios. A critical component of this program is identifying specific learning, career, and identity trajectories for students to adopt. Indeed, Squire argues that in one classroom, the most important learning was not any particular historical insight or media skills, but rather the realization that their indigenous literacy practices (e.g. using instant messenger to contact colleagues, using Google.com to look up unknown information) were core to life in an information-rich knowledge-based economy, even if they were prohibited in school.

Jenkins and colleagues (in press) use competitive game design workshops as another model for creating digital games literacy. Working with SONY Imageworks and other game and media programmers, Jenkins describes a program where students are given the task of identifying an existing media property, creating a game design and pitch materials, and then doing a 20 minute design brief before a team of experts from academia, industry, and business. Existing properties are chosen because they reduce the stress of writing an original story, minimize the dominance of one team member, and encourage a close analytic read of the property and rethinking how that property would work as a game. Students listen to lectures from inside and outside the industry on contemporary trends, the design of effective game characters, and market research. Groups are interdisciplinary and structured to include participants with media, business, and design expertise. Core to this project is the students' real-life approximation of an actual "pitch" session, so that the criteria for an effective project are aligned closely with those institutions outside of school (for profit game companies, non-profits, foundations).

These three sets of examples are but three examples of games literacy projects that attempt to blend close, critical readings of games with transitioning students from thinking as media consumers to media producers. Combined, they pose interesting questions for media literacy: How far do we go in insisting that students understand "the production" of media? How many English teachers would be considered *literate* in writing, if understanding the economics, politics, and social ramifications of book or newspaper publishing were considered a prerequisite to teaching English? Further, such projects, by attempting to tie to the commercial games industry may risk focusing on particular issues (profitability, market share) while overlooking others (under-represented voices). Regardless, these studies also emphasize that the biggest challenge in developing a critical vocabulary around games may not be with students, but with teachers, parents, and administrators who treat games as trivial, rather than influential cultural artifacts and practices.

Games as designed contexts for studying learning. Although there is a long tradition of games in education (particularly instructional design), historically there has been very little useful research into the effectiveness of games and simulations in instruction, with common critiques being that games research has lacked a coherent theoretical focus (Gredler, 1996; 2004; Rieber, 1996). One of the few fruitful paths of research has been psychological studies by Mark Lepper and colleagues, who used games as sites for studying highly motivated learning. Malone (1981) and Malone and Lepper (1987) examined games to build a theory of intrinsic motivation, finding that "good" games create highly motivated learning through challenge, curiosity, control, fantasy, competition and collaboration. Cordova and Lepper (1996) built prototypes of educational games in mathematics that leverage these properties in order to increase motivation, finding that more highly motivated students performed better than their peers on similar mathematics tasks. These well designed studies suggest how design-based approaches can lead to fundamental insights into cognition that improve the quality of instructional materials.

Building game-based learning environments. The most overt attempt at designing learning environments based on the technologies, learning principles, and social values of games to date has been MIT's Microsoft-funded Games to Teach Project. The Games to Teach Project (2003) was grounded in the premise that for too long educational technologists have treated computers

as technology rather than as media, thereby missing opportunities for designing powerful learning environments in the process. Holland, Jenkins, and Squire (2003) describe their 15 conceptual prototypes as ideal sites for generating and testing game theory, using prototypes as sites for identifying and exploring new avenues for game theory. These prototypes range from historical role playing to physics simulation games to massively multiplayer engineering games, each of which seeks to both explain how games work and to argue for new models of games.

One such game, *Supercharged!*, situates the student in the role of piloting a spaceship that has the power to adopt the properties of a charged particle; this design is in keeping with research that found expert physicists reason by imagining themselves literally “inside” physics problems. Students work through a series of maze-based levels based on popular platform game conventions that are designed to build their conceptual understandings of electrostatics, starting with Coulumb’s law and moving through most of an advanced high school physics electrostatics curriculum (Squire, Barnett, Higgenbotham, & Grant, 2004). Squire and colleagues found that students not only developed better conceptual understandings of physics, but a better understanding of *why* representations in their textbooks look the way they do. Further, Squire et al. found that lower achieving students emerged as leaders in their classes and showed the highest gains on pre-post tests. Teachers interviewed believed lower achieving students outperformed high achieving students because the activities and assessments in the unit did not primarily involve reading, interpreting and producing text, unlike most school activities.

Similarly, Dede and colleagues developed *Riverworld*, an interactive 3D environment where students investigate a host of environmental problems in an early 20th century village. The purpose of *Riverworld* is to help low-achieving students perform better in science through complex inquiry-based tasks where they must identify problems, collect data, synthesize information, and generate conclusions based on their understandings. Dede and colleagues (2005) find that not only did students learning in *Riverworld* perform better than students in a control group, but attendance rates were higher, there were less disruptions in class, and teachers and students alike reported higher levels of motivation. Importantly, students showed improved understanding of scientific inquiry as a result of participating in the program.

In another effort to apply gaming principles to formal learning environments, Barab and colleagues (2004; 2005) designed *Quest Atlantis*, a 3D environment aimed at helping elementary and middle school students learn science more effectively. Just as games are designed according to the values of a particular media property, *Quest Atlantis* is designed according to social commitments including an affirmation of diversity, personal agency, social responsibility, environmental awareness, healthy communities, and compassionate wisdom. The tasks and challenges of *Quest Atlantis* are designed to foster these virtues, creating a liminal space where students can inhabit a world that acknowledges and fosters these values. Barab et al. (in press) argues that students participate in *Quest Atlantis* out of a desire to help others, which suggests that contemporary models of motivation have omitted altruism as a powerful source of learning and virtue for designing learning environments.

These studies point to the rich pedagogical potential of games, but also to some of the limitations of creating pedagogical programs based on game-based principles in formal learning environments. All of these projects, while well-funded, are very primitive compared to even the

most basic commercial videogames. Whereas role playing games are designed to be played over 40 or more hours, most of the “units” developed with classrooms in mind were forced to work within a traditional school environment where students have a dozen hours at most for learning about a particular topic or concept. When compared to a contemporary massively multiplayer game, where participants *average* 20 hours per week online, students have significantly less time to become involved and to develop identities and investments within these virtual environments. Last, while engaging, such environments frequently pale in design sophistication compared to commercial games. As a result, a number of games researchers have begun experimenting with using commercial games in classrooms, hoping to use such games as “attractors” for academic intellectual work.

Teaching and Learning with Commercially Available Entertainment Games. Despite the broad popularity of management simulation, historical strategy, and games like *Sim City*, there have been very few studies to date of such games being used in classrooms. In perhaps the first study of such a game being used in a classroom environment, Squire (2005e) created custom historical scenarios for the game *Civilization III*. Squire’s scenario was designed to build on the game’s underlying geographical / materialist simulation model in order to help low achieving high school students understand key processes in world history (c.f. Manning, 2003; McNeil, 1968). Squire found that the game created a liminal space for students to explore world history, leading to basic background conceptual understandings in history, as well as deeper historical understandings of broad causes in history, which Dunn (2000) has called the “patterns of change” model of world history. Students previously marginalized from the study of history found spaces for their identities and lived histories to enter historical narratives. In particular, they found exceptionally rewarding that the “Rise of the West” might be seen as a particular product of its geographic location and access to trade networks (and ensuing colonization).

Attempting to use complex games in school-based learning environments also poses substantial challenges, however. Squire (2005e) reports many of these challenges: A single game of *Civilization* can take up to twenty hours to complete. The interface is opaque to new users, as it employs conventions built up over generations of strategy game iterations. Many girls do not immediately read the game as a girl friendly play space, as games have been historically gendered male, as has game playing culture. Further, as research on games would suggest, the game privileges literacies (such as generating original interpretations of history) that schools devalue. As Squire describes, the game “changes the rules of the game” on students, and it can be difficult to establish entirely new learning cultures within a traditional classroom environment.

In 2005, Egenfeldt-Nielsen conducted a similar study in Denmark, using *Europa Universalis* as a tool for studying European history. Egenfeldt-Nielsen reports implementation obstacles similar to Squire, as well as noting some conceptual problems in experiential game-based learning approaches. These conceptual problems include a mismatch between learning outcomes through games and state curricular goals, time constraints in setting up games, differential learning outcomes across students, and the complexity of game interfaces for novices. Egenfeldt-Nielsen describes learning through game-based environments as a hermeneutic process whereby students observe phenomena, interact with game rules, and draw interpretations based on previous understandings of history, just-in-time information provided by

teachers and outside resources, and experiences in the game environment. As such, game-based learning in intentional learning environments is a hermeneutic process that resembles learning in a research lab. Learning is driven by a combination of students' questions, emerging experiences and findings, and historical interpretations.

Based on these findings, Squire and colleagues have been examining the potential of *Civilization III* and *Civilization IV* in after-school environments (Squire, Giovanetto, DeVane, & Durga, 2005). Whereas previous game development and curricular enactment programs have struggled to accommodate the demands of schools, formal after-school environments offer intriguing possibilities for rethinking educational practice for a digital age. Squire and his colleagues' program engages students iteratively in building and playing historical "mods" or scenarios for *Civilization III* and *Civilization IV*. They report the processes by which students learn to build these historical mods as a process of: (1) developing interest in the game, (2) developing advanced knowledge of the game systems, (3) seeing iterative models of the game tailored in response to students' play, (4) adapting scenarios to enhance their play, in particular, to create games for other students to enjoy, and (5) building models to highlight particular game dynamics, such as the domination of the Vikings in medieval Europe. Squire and colleagues are finding that in controlled environments, students can and do develop literacies as game producers, which is marked by a systemic level understanding of game systems and design. They posit that such literacies could be developed in classrooms, but would require systemic changes that give teachers and students more freedom in choosing what they study and more flexibility within the curriculum.

Conclusions

A recent *New York Times* article that describes the rise of academic programs in higher education, quotes former Senator and current president of the New School, Bob Kerry, as arguing that "The skills and methods of videogames are becoming a part of our life and culture in so many ways that it is impossible to ignore," (quoted in Schiesel, 2005). This suggests an alternative narrative to the predominant broadcast media story that "games are bad for you." Each year that the Nintendo generation ages, the average age of gamers rises, and some reports suggest that aging Generation Xers and baby boomers are turning to games as a way to stay social when there are children in the house (you may not be able to go out every night, but you can log into *World of Warcraft*), or ward off the effects of aging (Sheff, 1994; Oblinger & Oblinger, 2005; Steinkuehler, 2005). As games become increasingly mainstream, literacy scholars are looking to videogames as sites for studying emerging digital literacies.

Perhaps the alarm bells sounding the decline of "reading" do not signal a "loss" of literacy, but the emergence of a new constellation of literacy practices that could someday transform our social institutions. If the history of media holds true, videogames, like film or television, are not going to replace books but live along side them (cf., Thorburn & Jenkins, 2004). Games are becoming ingrained within a range of political, military, commercial, and – at least for now on the margins of – educational systems, bringing with them attendant changes in cultural practices. Cases like *America's Army* show how a game made for military recruitment purposes operates differently from a 30 second recruiting advertisement run during a televised football game, which, in turn, changes what it means to brand and recruit new military personnel.

Similar examples can be found in politics: (e.g., *The Mass Balance* game where citizen-players are challenged to try and balance the budget), advertising (e.g., YaYa Media's *Jeep* driving game, where sales people and consumers can test drive new vehicles), or military training / entertainment (e.g., *Full Spectrum Warrior*, a game that challenges you to lead squadrons through urban militarized zones) (Gee, 2005; Squire, in press). These examples suggest how emerging literacies are about more than physical (or digital) materials; they are about emerging cultural practices that reciprocally transform people and social institutions.

Perhaps not surprisingly, schools have been slower than the military and industry in taking advantage of the opportunities in games. The literacies that schools are based around include "mastery" over pre-determined, increasingly federally-mandated content, reading and writing traditional school based genres of text (i.e. The 5 paragraph essay, the term paper), ritualized performance on decomposed tasks, participation in activity cleaved from other social processes and institutions. Game-based literacies include a constellation of literacy practices that are quite different: texts are spaces to inhabit, learning as a productive, performative act, knowledge is legitimized through its ability to function in the world, participation requires producing as well as consuming media, expertise means leveraging digital spaces to further one's goals, and social systems have permeable boundaries with overlapping trajectories of participation.

As games become more culturally entrenched, the idea of using games in education may be passing from an *opportunity* to an *imperative*, if we are to create an educational system that adequately prepares students for life in an information/ knowledge rich economy. Writing in late 2005 on the day of the release of the Xbox 360, I would claim that games are only rising in cultural prominence and acceptance as SONY, Microsoft, and Nintendo, and Electronic Arts all vie to expand their markets beyond traditional gamers, and seem poised to become even more mainstream. At the same time, schools are being regarded by students as less relevant than ever; for the first time in the United States' history, for example, a majority of students – even the successful ones – see school as little more than credentialing spaces (Baines & Stanley, 2003). In schools, we label those students who exhibit the qualities of gamers – increased independence, preference for multi-tasking, decreased respect for authority – as ADHD; outside schools, these same students are championed as leaders of the new economy (Gee, Hull, & Lankshear, 1996). The history of education also suggests that if these emerging literacies do prove to be foundational to success in the new economy, upper-middle class parents will make sure that their children develop these skills, and in particular, have access to not just games but the affinity spaces and self-organizing learning systems that are crucial to becoming producers of and with digital media.

The real question will become how will underfunded and overlooked urban schools respond.

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